

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

1. Calculate the heat required to raise 1 L of water from 30 Â°C to 60 Â°C :

Volume of 1 L = 1,000 g (1 g/mL for water)

$Q = mc\Delta T$ (heat = mass Ã specific heat Ã temperature change)

$\Delta T = 60\text{Â}^\circ\text{C} - 30\text{Â}^\circ\text{C} = 30\text{Â}^\circ\text{C}$

$Q = (1,000\text{ g}) \times (1\text{ cal/g}\text{Â}^\circ\text{C}) \times (30\text{Â}^\circ\text{C}) = 30,000\text{ cal} = 30\text{ Kcal}$

2. Calculate the heat required to raise 1 kg (1,000 g) of aluminum from 30 Â°C to 60 Â°C :

$Q = (1,000\text{ g}) \times (0.215\text{ cal/g}\text{Â}^\circ\text{C}) \times (30\text{Â}^\circ\text{C}) = 6,450\text{ cal} = 6.45\text{ Kcal}$

So, the answer is (G). 30 Kcal , 6.45 Kcal

</think>< answer> G